

Optimizing Airbnb's Performance in Copenhagen: Data-Driven Insights for Growth & Success

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Project Overview

Key Stakeholders
Executive Management
Operational and Business Development Teams

Goals

- Analyze and identify critical success factors & trends
- Deliver strategic insights
- Provide actionable recommendations to
  - enhance performance
  - maximize revenue
  - drive sustainable growth
  - utilize new opportunities

Questions

What factors are driving guest satisfaction?

What impact does superhost status have on host performance?

Which neighborhoods offer the greatest opportunities for growth and profitability?

...

Measures

Overall market dynamics

Host performance metrics
- Response time
- Superhost status
- Average ratings

Neighbourhood metrics
- Average price
- Availability trends
- Number of ratings

Data Cleaning & EDA



Data cleaning

Exploratory data analysis

Libraries:
- Pandas
- Numpy
- Matplotlib
- Seaborn

Data storage

Central database

Real-time access

PgAdmin

Visualization

Dashboards

Design concept:

- Defining target audience
- Following Shneiderman's mantra

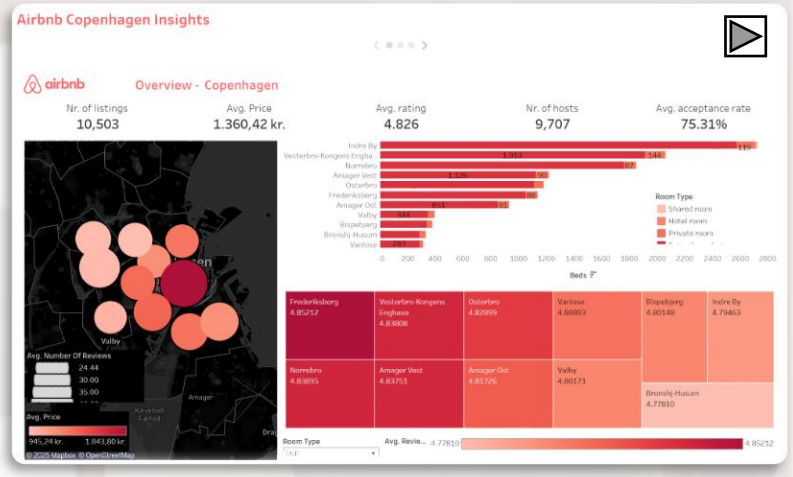
  „Overview first, zoom and filter, then details on demand"
- Hierarchy, top-down
- Clean and branded colour scheme

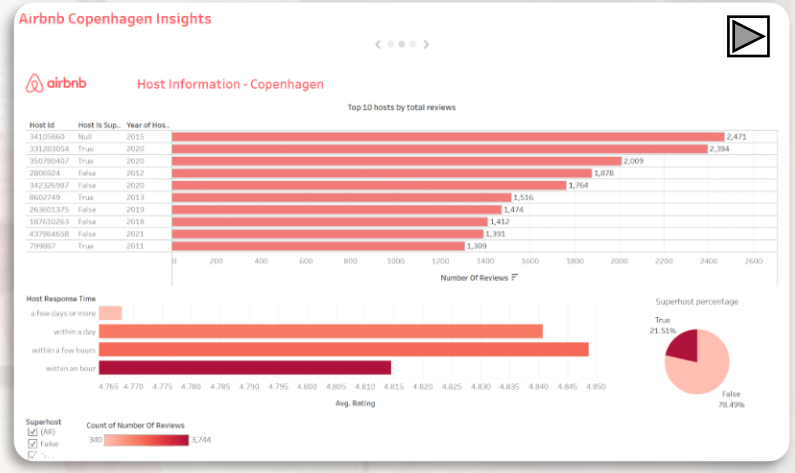
Key Features:

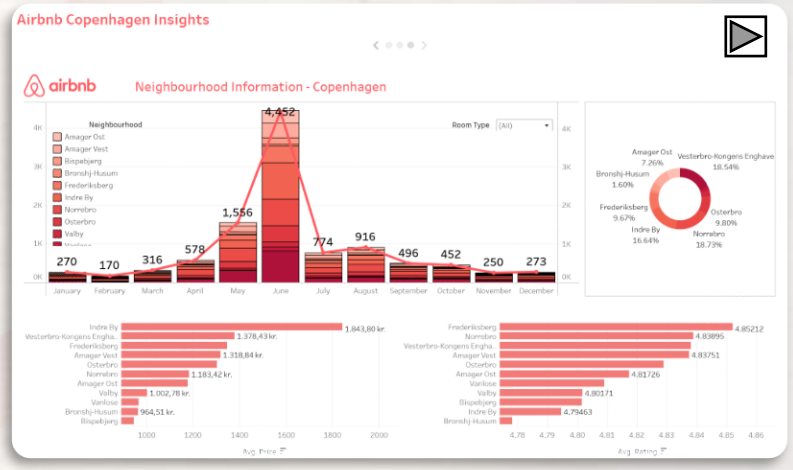
- Dynamic connection to PostgreSQL for real-time updates
- Interactive filters for granular insights
- Hover-over tooltips

Visual elements:

- Bar charts
- Pie charts
- Scatterplot & geographical map
- Histogram
- Treemap & heatmap







https://public.tableau.com/app/profile/meng.yun.lu/viz/Airbnb_dashboard_17354801209280/Story1

Limitations & Conclusion

Limitations

- Data Denormalization and Redundancy
- No Formal Schema Design
- Static Dataset
- Analytical Scope

Conclusion

- The project successfully identified critical success factors and provided actionable insights
- Dashboards empower stakeholders to make data-driven decisions, enhancing performance and driving growth

Future Improvements

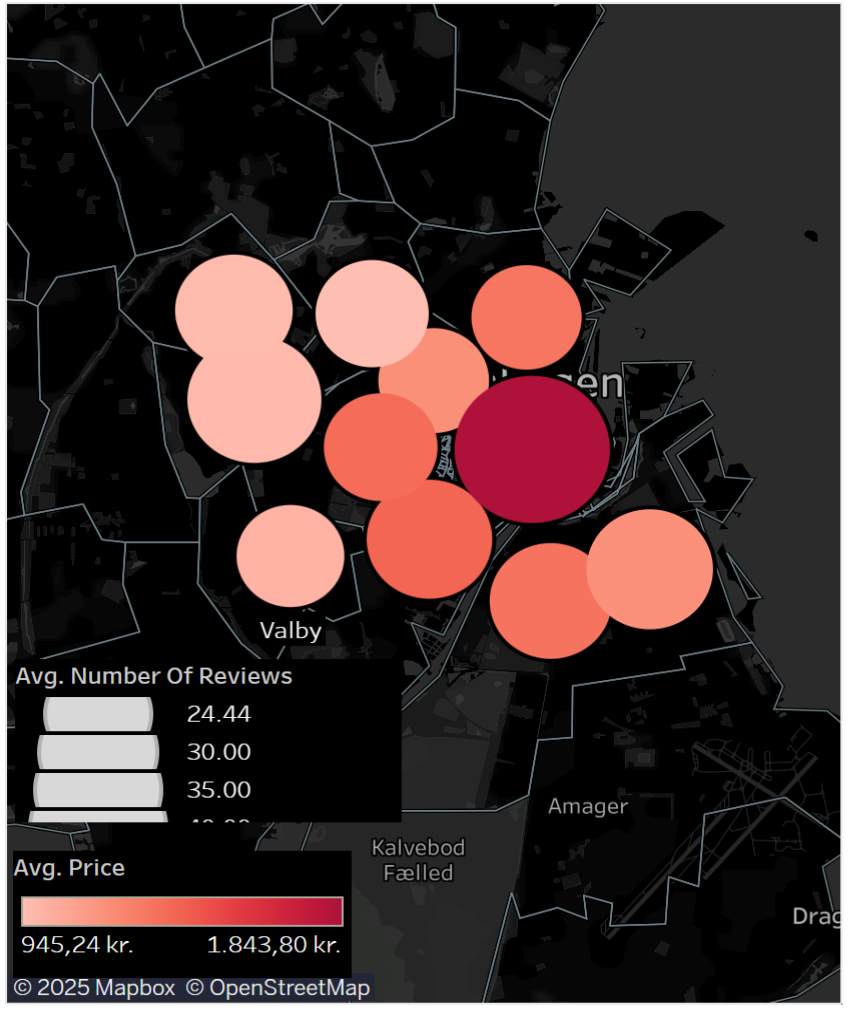
- Reduce redundancy
- Improve performance
- Implementing a Star Schema
- Incorporate real-time or regularly updated data

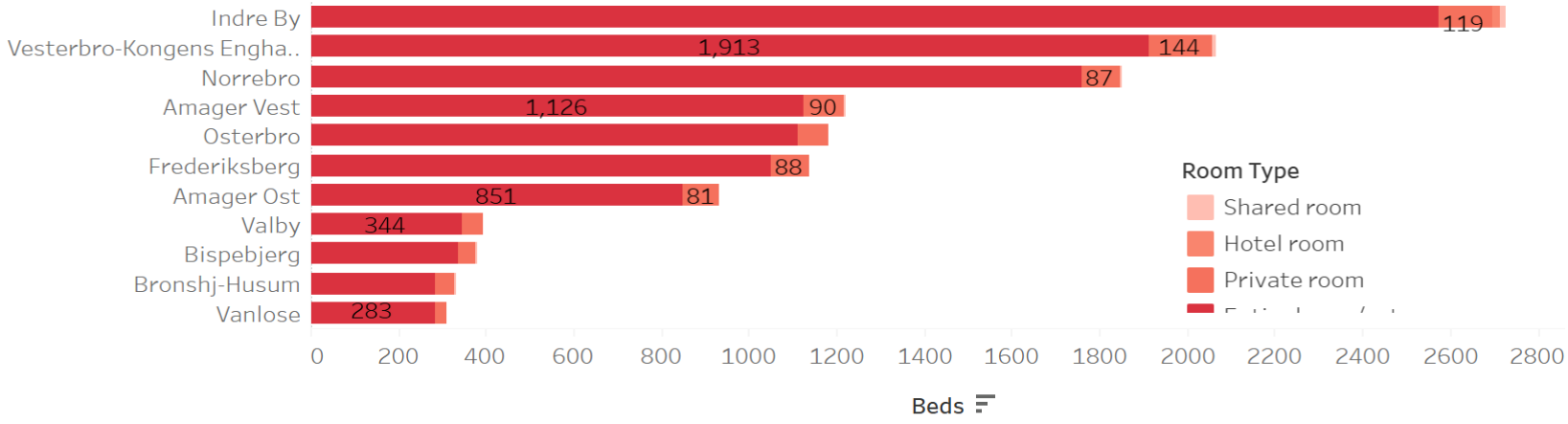

Thank You
for your
attention!

Airbnb Copenhagen Insights

Overview - Copenhagen

Nr. of listings	Avg. Price	Avg. rating	Nr. of hosts	Avg. acceptance rate
10,503	1.360,42 kr.	4.826	9,707	75.31%







Room Type
(All)

Avg. Revie... 4.77810 4.85212

Airbnb Copenhagen Insights

Host Information - Copenhagen

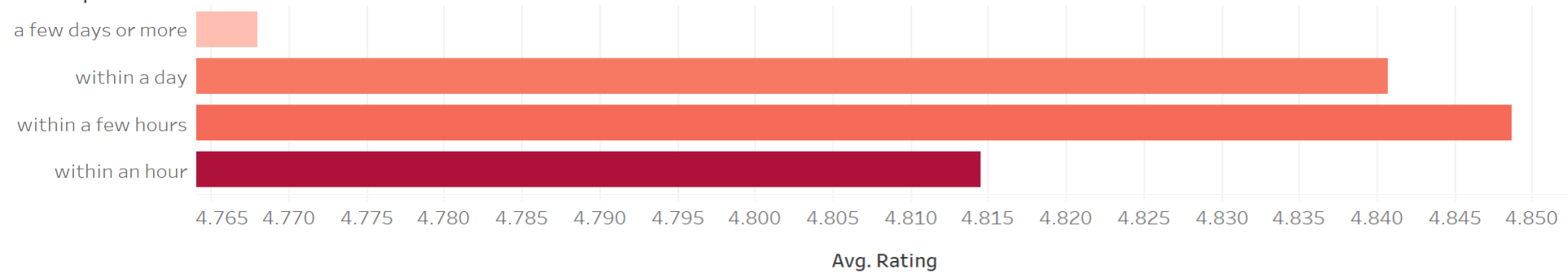
Top 10 hosts by total reviews

Host Id	Host Is Sup..	Year of Hos..	Number Of Reviews
34105860	Null	2015	2,471
331283054	True	2020	2,394
350780407	True	2020	2,009
2806924	False	2012	1,878
342326987	False	2020	1,764
8602749	True	2013	1,516
263601375	False	2019	1,474
187610263	False	2018	1,412
437864658	False	2021	1,391
799887	True	2011	1,309

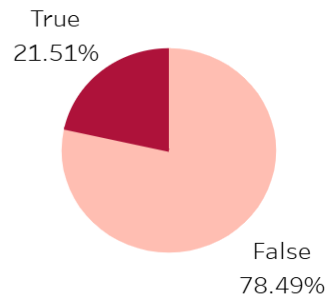
0 200 400 600 800 1000 1200 1400 1600 1800 2000 2200 2400 2600

Number Of Reviews

Host Response Time



Superhost percentage



Superhost
- [x] (All)
- [x] False
- [x] True

Count of Number Of Reviews

340 3,744

Airbnb Copenhagen Insights

Neighbourhood Information - Copenhagen

